generic_csr — Design Specification

`generic_csr` is a small control/status register block with a single-port CSR interface. It is parameterized by `ADDRW` and `DATAW` (both default to 8, minimum 8). All logic is clocked on the rising edge of `clk`; `rst_n` is an active-low asynchronous reset. An access is accepted only when `chip_enable` is high. A write (`write_enable`) updates the register on the next rising edge. A read (`read_enable`) returns data on `data_out` one cycle later; `data_out` holds its value until the next read and is cleared by reset. `write_enable` and `read_enable` must never be high in the same cycle.

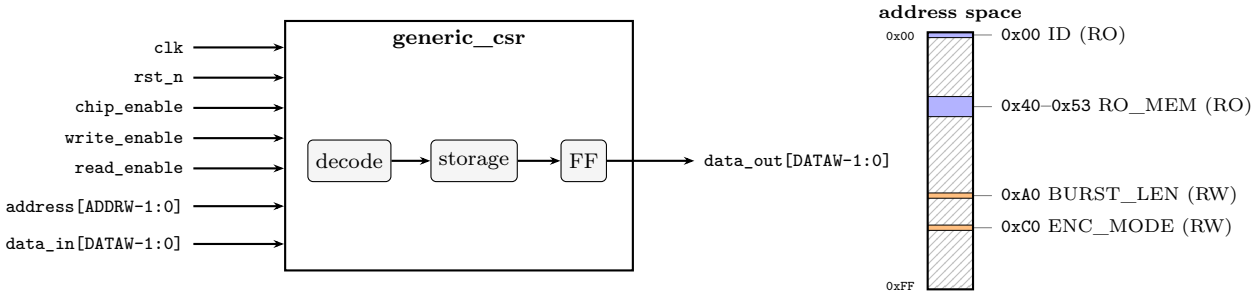

Figure 1: Interface of `generic_csr` (left) and its address space drawn to scale (right). Read data passes through an output register (FF), giving one cycle of read latency. Blue: read-only; orange: read/write; hatched: unmapped.

Table 1: Address map. Unlisted addresses read as 0 and ignore writes.

Address	Name	Access	Reset	Description
`0x00`	ID	RO	`0xBE`	Constant identifier; writes ignored
`0x40`–`0x53`	RO_MEM	RO	`0x00`	20 entries; cleared only by `rst_n`; writes ignored
`0xA0`	BURST_LEN	RW	SHORT	SHORT, AVERAGE, LONG
`0xC0`	ENC_MODE	RW	PLAIN	PLAIN, HASH, ECC
other	—	—	—	Reads return 0; writes ignored

Writing any other value to an RW register is ignored and the register keeps its previous value.